

PRUDHVI MUNAGANTI

Mechanical Design Engineer | Automotive Exterior Design & Surface Modelling

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PROFESSIONAL SUMMARY

Mechanical engineer with two years of hands-on experience in design and production, now pursuing a Master's degree in Sustainable Technology Management (Automation, Robotics and Industry 4.0) at SRH University of Applied Sciences, Berlin. Over the past year, I built self-directed expertise in automotive reverse engineering, taking STL scan data through to finished CAD surfaces in CATIA V5, and am now extending this into Autodesk Alias AutoStudio to develop Class-A surfacing fundamentals: curve quality, patch layout, surface continuity, and reflection analysis. Currently based in Berlin, proficient in English with basic and actively improving German, seeking a mandatory internship required as part of the Master's programme, ideally in automotive exterior design, CAD, or surface modelling, to apply this foundation professionally and continue developing under experienced surfacing engineers.

CORE COMPETENCIES

- Reverse engineering from STL mesh and 3D scan data
- Surface reconstruction and reverse-engineered CAD modelling
- CATIA V5 (STL import, axis alignment, 3D curve creation, surfacing, solid conversion)
- Cross-functional teamwork and communication
- Autodesk Alias AutoStudio 2027 (Developing curve creation, patch planning, surface construction, Class-A surfacing fundamentals)
- Reflection analysis, deviation analysis, surface and curve continuity checks
- CAD modelling in Autodesk Inventor, Fusion 360 and Siemens NX
- Familiarity with Siemens Simcenter for digital twin applications
- Strong spatial reasoning, geometric accuracy and independent problem-solving

PROFESSIONAL EXPERIENCE

Designer, Quality and Production Engineer — **August 2023 – March 2025**

Turbo Engineering Services — **Hyderabad, India**

- Contributed to mechanical component design using CATIA and Autodesk Inventor, developing part models and technical drawings within a production-oriented engineering environment
- Performed quality checks on manufactured parts, helping maintain output consistent with workshop standards
- Worked across design, assembly and production stages, developing a practical understanding of how components move from CAD model to finished part
- Gained exposure to lean, efficiency-focused workshop practices through day-to-day production support

INDEPENDENT AUTOMOTIVE SURFACING & REVERSE ENGINEERING PROJECTS

Self-directed practical development, undertaken independently alongside academic studies

- Reconstructed full-vehicle and body-panel surfaces (including a Toyota Supra, an Alfa Romeo 155 V6 body side, a Mustang front hood and fender, and multiple fenders and spoilers) from STL scan references in CATIA V5, applying axis alignment, 3D curve creation, patch-based surfacing and multi-section surface techniques
- Developed airtight, gap-free closed surface models, validating the resulting geometry through deviation analysis and preparing it for solid generation
- Developed structured patch layouts and maintained continuous surface flow across complex panels, including wheel arches, headlights and character lines
- Transitioning core reconstruction skills into Autodesk Alias AutoStudio, applying STL mesh preparation, curve creation, patch planning and reflection-based evaluation to build Class-A surfacing fundamentals
- Developed custom reference curves rather than relying solely on direct mesh tracing to better capture underlying design intent
- Built this portfolio independently, without formal automotive employment, as direct preparation for a first professional role in automotive exterior design

EDUCATION

Master of Engineering (M.Eng.), Engineering and Sustainable Technology Management — **Currently Enrolled**

SRH University of Applied Sciences — **Berlin, Germany**

Focus: Industry 4.0, automation, robotics, and 3D manufacturing

Bachelor of Engineering (B.E.), Mechanical Engineering — **June 2019 – March 2023**

Velagapudi Ramakrishna Siddhartha Engineering College — **Vijayawada, India**

CERTIFICATIONS

National Cadet Corps (NCC) Certificate, leadership and discipline